

InstructFX2FX: **Text-to-Preset** Generation with LLM Initialization and **Human-in-the-Loop** Gradient Refinement

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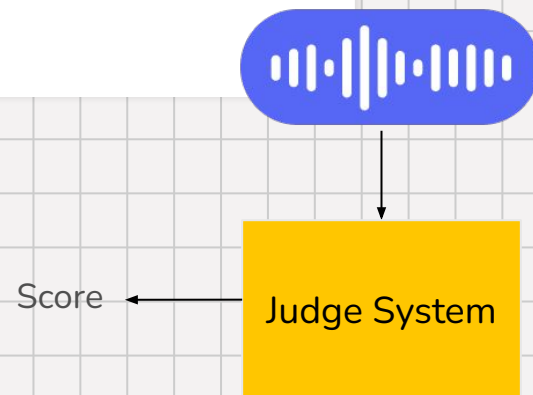
Question Statement

Abstract

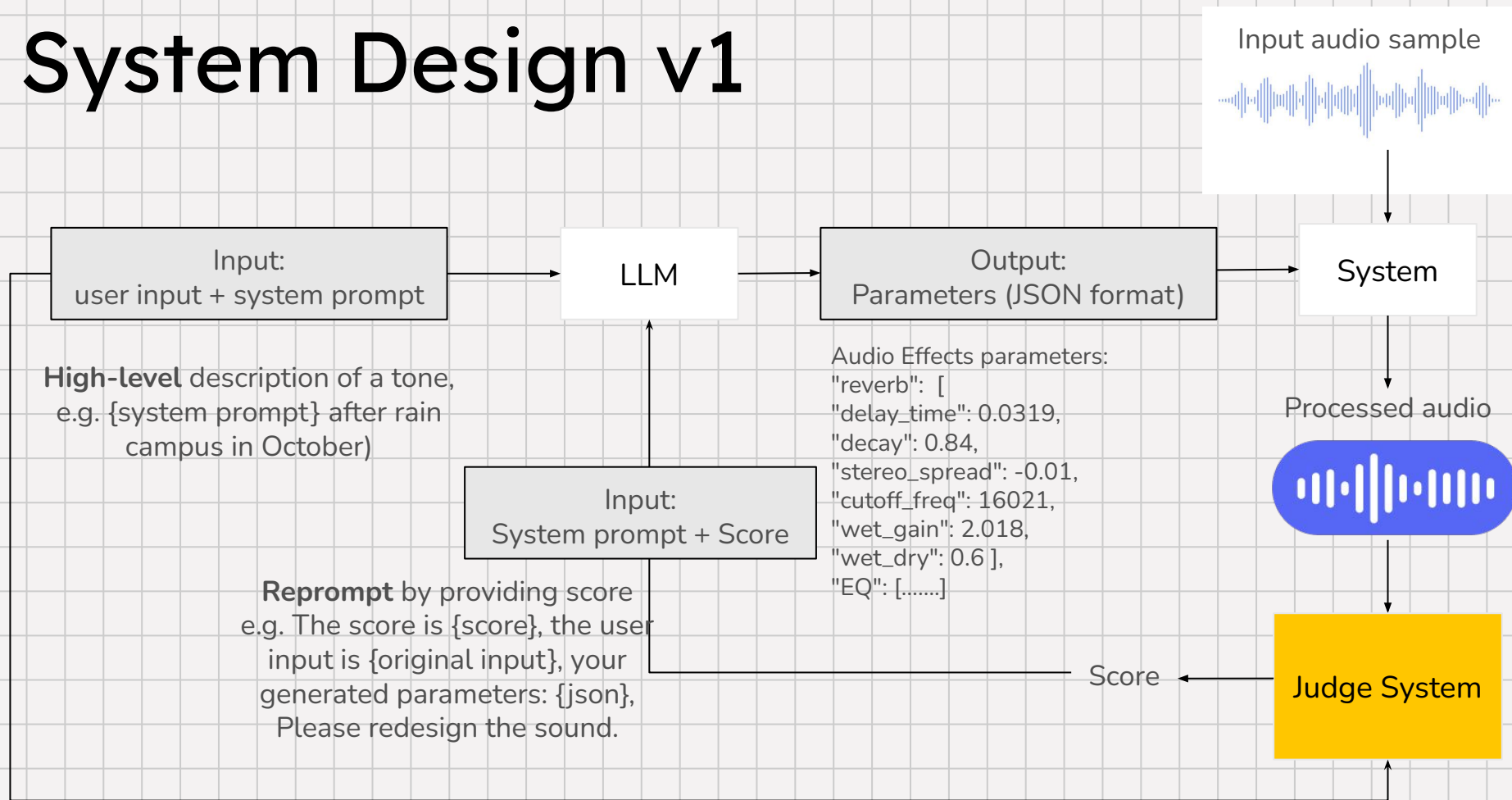
- Nowadays, LLMs and generative audio models can already produce tones/music from text. *(Can Large Language Models Predict Audio Effects Parameters from Natural Language?)*
- But there is still (1) no reliable way to **evaluate** whether those tones actually match user intention or its quality; (2) there is no robust way to **refine or adjust** the sound.

Research Abstract

- Create an fair **evaluation system** for tone/music
- **Refine** mechanism for editing the generated tone/music
(re-prompting, human-in-the-loop...)



System Design v1



Case study - warm.wav



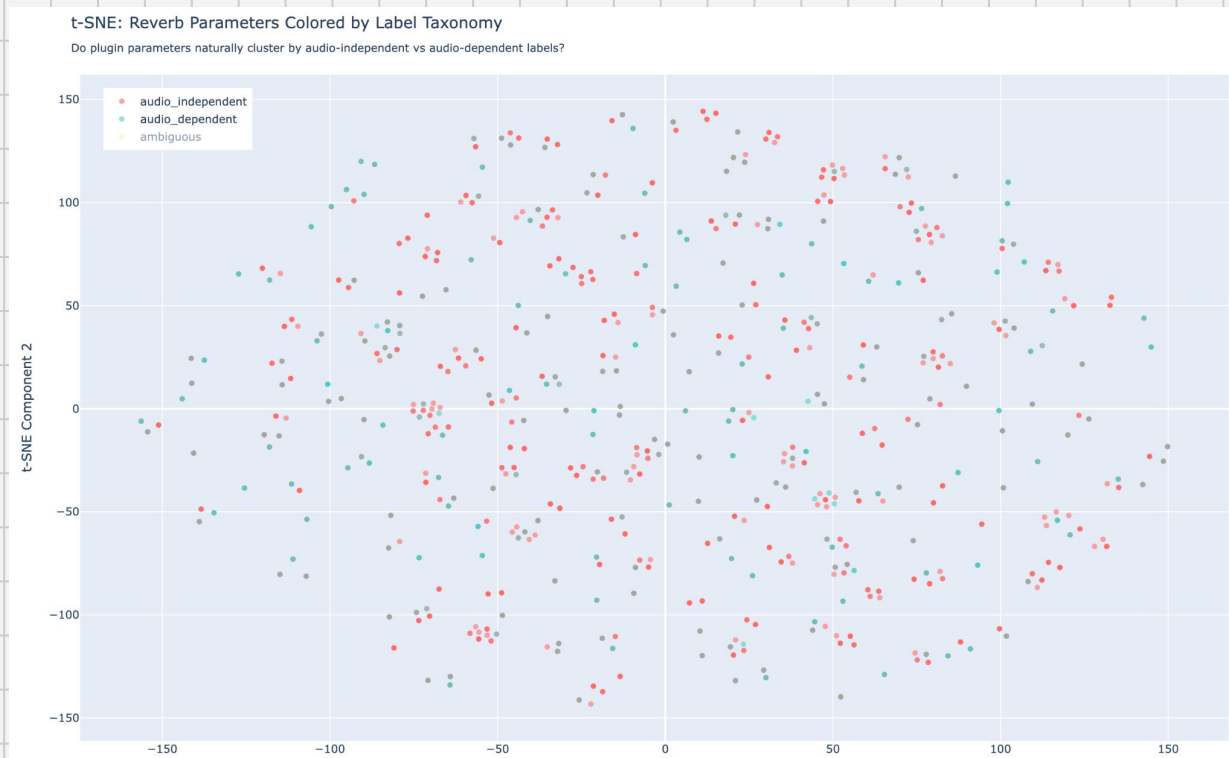
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SUMMARY COMPARISON
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Rank	CLAP	MERT	Hybrid
1	church-bell	high-bass	classy
2	big-church	classy	low-fidelity
3	cathedral	low-fidelity	high-bass
4	cathedral-like	engulfing	chambers
5	chaotic	roomy	old-organ
6	church-like	wavy	old-timely
7	glorious	high-pitch	waved
8	ominous	large-auditorium	dominant
9	chambers	liquid	mixed
10	classical	comfortable	loving

Maybe deep into the data.....?

Tsne on SocialFX? graphs?

descriptors are not
separable!



Problem & Challenges of v1

1. It's hard to provide a fair “score” in the judge system:

- *Semantic descriptors in SocialFX are extremely noisy.*
- *No standard answer for what “warm” or “airy” should be.*
- *How to quantize music into a score?*

Human-in-the-loop!

→ Reinforces that our judge system lacks semantic grounding.

2. Reprompting may NOT converge

Even if the judge outputs a score like 3/10 ($<$ threshold) and we reprompt the LLM:

“Given your score is 3/10, please redesign the sound.”

→ LLM reprompting is fundamentally unstable.

Optimization Problem!

Question Statement

Abstract

- Nowadays, LLMs and generative audio models can already produce tones/music from text. *(Can Large Language Models Predict Audio Effects Parameters from Natural Language?)*
- But there is still (1) no reliable way to **evaluate** whether those tones actually match user intention or its quality -> **FAILED? Fine! Just let user to tell how they feel!**
(2) there is no robust way to **refine or adjust** the sound.

Question Re-Statement

Instead of asking:

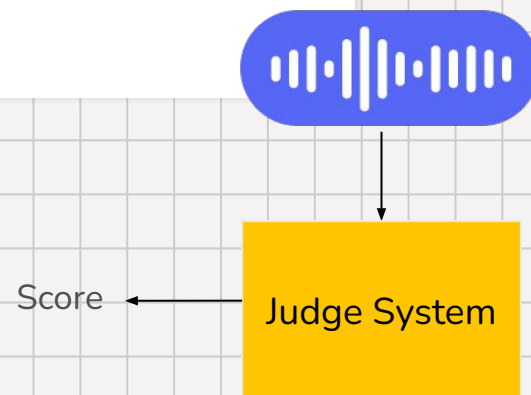
“How do we ***judge*** a sound?”

We should ask:

“How do we ***move*** a sound toward the user’s intention?”

Research Abstract

- Create an fair **evaluation system** for tone/music -> **Human-in-the-loop!**
- **Refine** mechanism for editing the generated tone -> **Optimization!**



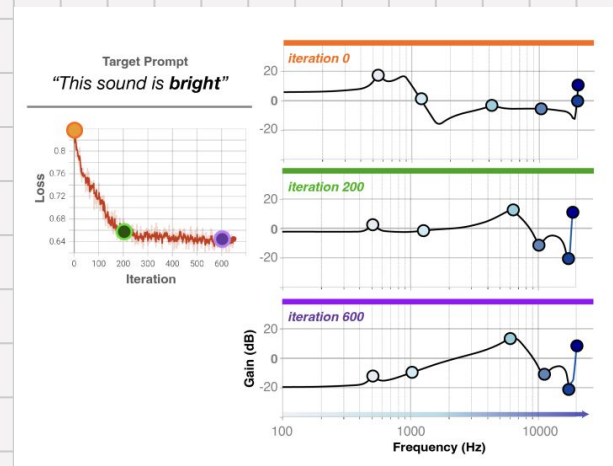
System Design v2: Human-in-the-Loop Optimization

Related Works

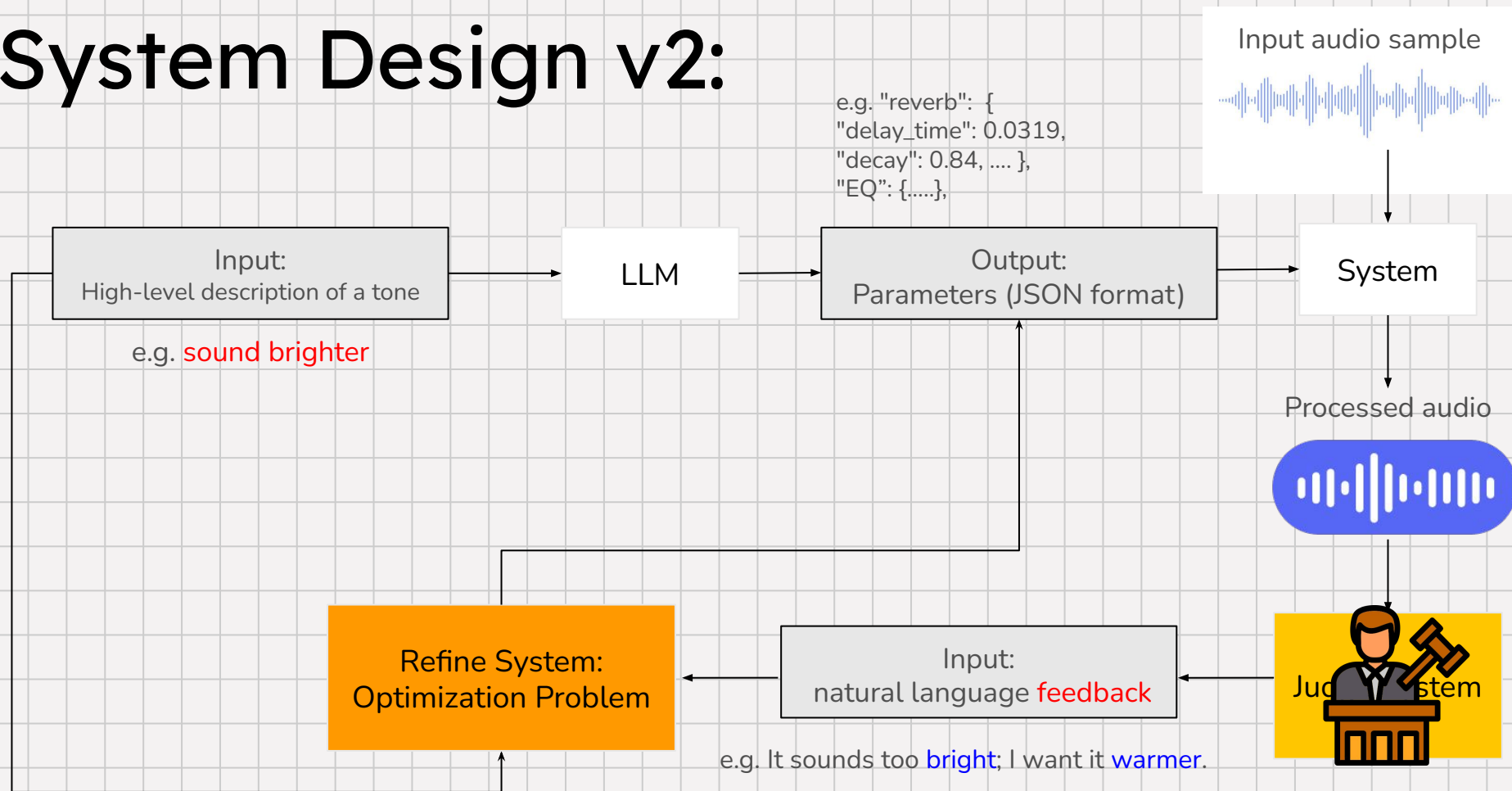
Inspired by *Text2FX*, *InstructPix2Pix*:

- *Text2FX* **optimizes** parameters toward a prompt
- *InstructPix2Pix* allows instruction-based refinement

We propose a **tone-editing system**, not just a tone generator.

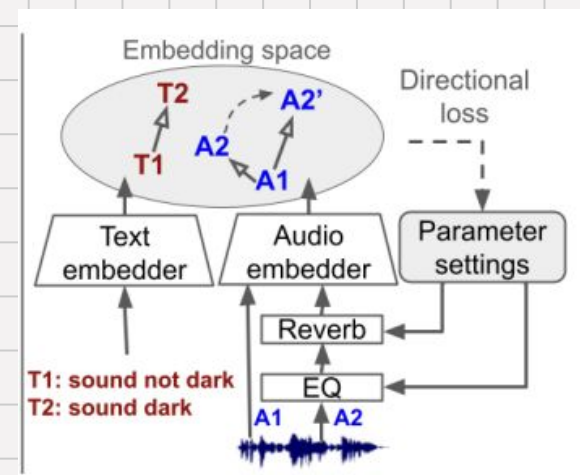
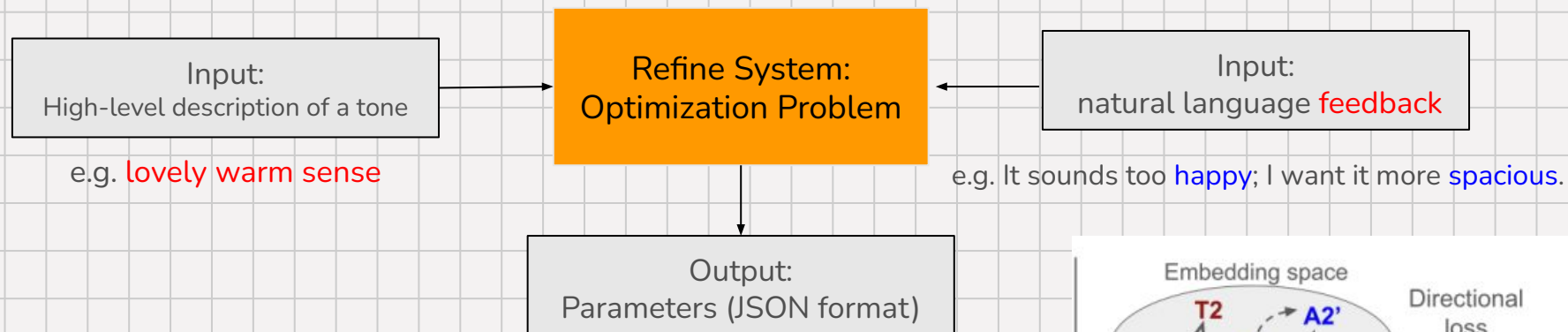


System Design v2:



Refine System:

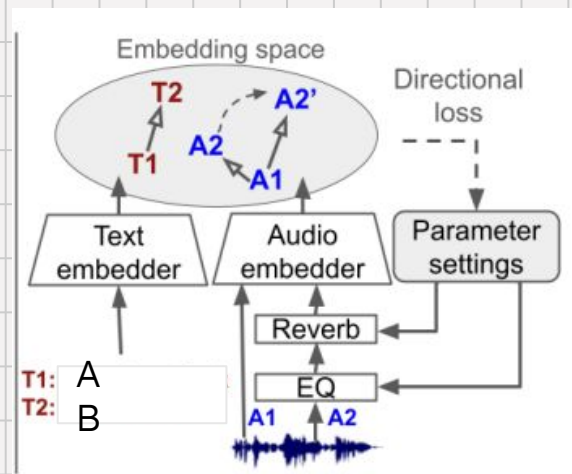
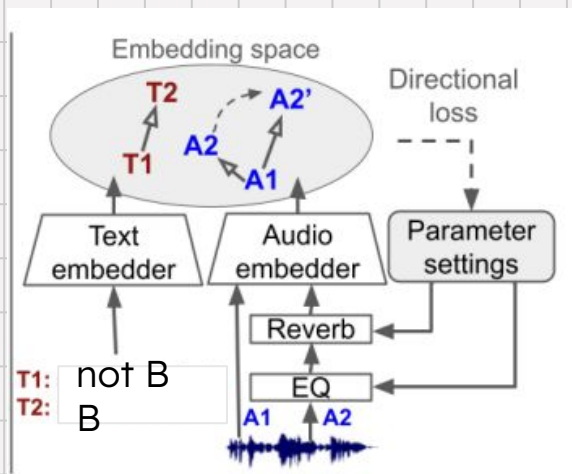
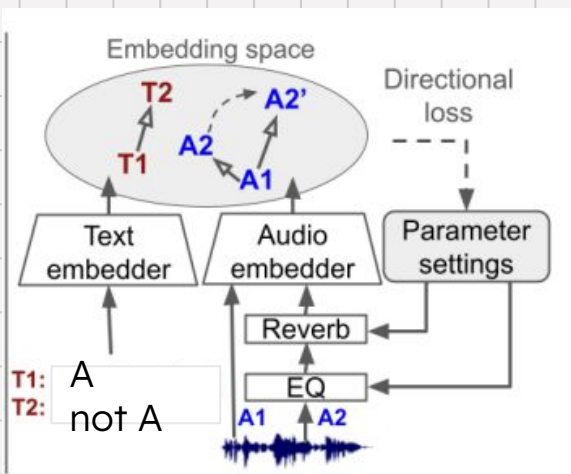
This avoids the need for a **hard judge** and solves the **convergence issue**.



Experiments:

Initial Prompt: This is a piano music. I want it be brighter

Reprompt template: It sounds too A; I want it more B.



Results:

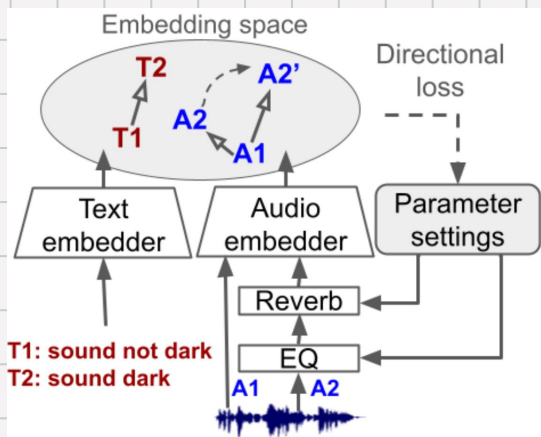
InstructFX2FX preserves the LLM's flexible, semantically meaningful starting point, but add a smart refinement system that makes the sound really editable.

-> The demos show it is far more effective than simply re-prompting the LLM.

Limitations:

- LLM prompt still requires tuning
- the optimal number of refinement epochs and whether to use early stopping remain open questions.
- The convergence time is relatively slow (several seconds), which may be impractical for real-time plugin usage.

Refine System Limitations



Limitations:

1. T2 - T1 does not reflect a robust “A” vs not-“A” axis.
2. Extra semantics in “not A” leak into the direction.
3. The resulting direction can be misaligned with perceptual attributes.

Possible approach: define attribute meaning using distributions rather than single prompts

$$S_A^+ = \{\text{prompts expressing } A\}$$

$$S_A^- = \{\text{prompts expressing the opposite of } A\}$$

$$\mu_A^+ = (1/|S_A^+|) \sum_{t \in S_A^+} f_{\text{text}}(t)$$

$$\mu_A^- = (1/|S_A^-|) \sum_{t \in S_A^-} f_{\text{text}}(t)$$

$$d_A = (\mu_A^+ - \mu_A^-) / ||\mu_A^+ - \mu_A^-||$$

Thank you!